

Setup Guide: Autonomous Line Tracking

Time Allocation: 90 Minutes

Challenge Overview

The robot must autonomously navigate a 2-meter black line on a white floor without human intervention.

Environment & Arena Setup

- **Surface Preparation:** Secure a smooth, matte white surface (such as poster board, butcher paper, or a vinyl floor) that measures at least 1 meter wide by 3 meters long to allow adequate clearance around the track.
- **Track Creation:** Using 1-inch to 2-inch wide black electrical tape or matte black painter's tape, create a continuous 2-meter line. Incorporate a mix of straight segments and gentle curves to test the students' navigation algorithms.
- **Start and Finish Zones:** Clearly mark the starting line and the stopping point on the track using contrasting tape or visual markers so students know where to initiate their autonomous runs.
- **Lighting Conditions:** Ensure the arena has consistent, ambient overhead lighting. Avoid direct, harsh spotlights or shadows cast across the track, as these can disrupt optical light sensors.